Beyond the Information Bound: End-to-End Co-Design of Sensor Hardware and Adaptive Measurement Policy

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Abstract

End-to-end optical inverse design co-optimizes continuous physical hardware with a *one-step* learned reconstruction, with a growing literature explicitly adopting mutual-information or Fisher-information rewards as the training objective [11, 52, 53, 76]; adaptive-sensing POMDP theory optimizes a *multi-step adaptive policy* on fixed hardware. Neither addresses what a real reconfigurable sensor is: a physical geometry whose operator reacts to data as it arrives. We close this gap. Using the pointwise expected information-gain functional $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ as a family of compatible scalar rewards, we formulate the joint optimization of continuous hardware $\boldsymbol{c}$ and Bellman-optimal adaptive policy π end-to-end. An envelope-theorem identity removes the policy Jacobian from the geometry gradient, making the outer loop tractable even when the inner Bellman solver is approximate. A principled relaxation hierarchy, exact dynamic programming, deep reinforcement learning, certainty-equivalent control, myopic Bayesian D-optimal design, and extended-Kalman-filter Gaussian-trace surrogates, renders the formulation portable from 16-state discrete POMDPs to 10^5-pixel photonic topologies. Three case studies demonstrate the framework at three scales and under three different choices of Φ: (i) a radar beam-search POMDP solved by exact enumeration where the classical information-bound-guided geometry loses a factor of $2.8\times$ in attainable adaptive value to the joint-DP optimum, (ii) a superconducting-qubit flux sensor in which joint dynamic-programming co-design achieves a statistically significant 13.4% reduction in deployed mean-squared error ($z = +11.5\sigma$) over the joint Bayesian Cramér–Rao (PCRB) baseline, and (iii) a photonic metasensor with 90,000 free design pixels co-optimized via the Bayesian Fisher-information-matrix surrogate. We show that the central operational claim, joint $(\boldsymbol{c}, \pi)$ optimization strictly outperforms the decoupled classical blueprint, holds provided three engineering preconditions are met: a sufficiently rich action grid, a scalar reward aligned with the deployment metric, and a physically realistic hardware box. We advance the Expected Value of Perfect Information as a useful diagnostic pre-screen, not a go/no-go gate.

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1 Introduction

Sensors are a concatenation of a physical front-end (a lens, an antenna aperture, a superconducting circuit, a diffractive metasurface) and a measurement protocol that interrogates the front-end and reconstructs a quantity of interest $\boldsymbol{x}$ from noisy outputs $\boldsymbol{y}$. Two largely separate research communities have each co-designed *half* of this pipeline. On the hardware side, the last decade of computational inverse design has established that the physical geometry $\boldsymbol{c}$ of an optical element can be optimized end-to-end with a downstream reconstruction, typically a fixed, feed-forward neural network [3, 19, 39, 51, 65, 97, 108**?**]. Inverse-designed photonics and metaoptics now routinely reaches the electromagnetic performance bounds of the underlying scattering problem [23, 32, 48, 60–62, 70, 113]. On the policy side, the adaptive-sensing, sensor-management, and sequential-Bayesian-experimental-design communities have established a mature toolkit for constructing *multi-step* measurement policies that adapt to incoming data, but they invariably assume the physical hardware is fixed [1, 18, 21, 40, 41, 43, 49, 56, 57, 63, 94, 98, 99].

These two bodies of work address disjoint halves of the co-design problem. End-to-end optics is *non-adaptive* in time: the learned reconstruction takes a single-shot observation and returns an estimate; there is no concept of "what should I measure next given what I have already seen?" Adaptive sensing, conversely, treats the hardware as immutable. The missing piece is the joint optimization of *continuous* physical hardware with a *multi-step Bellman-optimal* adaptive measurement policy. Existing near-relatives, cognitive radar with MIMO waveform adaptation [4, 13, 35, 38], quantum metrology with Kitaev-style phase estimation on fixed circuits [14, 15, 17, 27, 34, 55, 78], programmable metasurface imaging with reconstruction-only adaptation [45, 117], and variational quantum sensing with trainable but fixed ansatze [30, 82], each either decouple hardware from policy or approximate the inner loop with a fixed-architecture neural ansatz that lacks a physical-geometry interpretation.

Closing this gap requires a formulation that (a) is cleanly defined across discrete and continuous state spaces, (b) admits a tractable outer gradient with respect to the physical geometry, and (c) provides a hierarchy of relaxations that scales from exact dynamic programming on small POMDPs to high-dimensional continuous hardware optimization. We supply all three.

1.1 Contributions

1. **Unified formulation.** We identify a *family* of pointwise scalar rewards $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ that fit the same joint-$(\boldsymbol{c}, \pi)$ Bellman machinery. Three canonical members are (i) the mutual-information-aligned Φ_{MI} following Lindley [63], (ii) the Bayesian mean-squared-error aligned Φ_{Var} tied to the deployment metric by the tower law of variance, and (iii) the Fisher-information aligned $\Phi_{\log J}$ following Cramér–Rao [26, 86]. Each reward picks out a different operating point; choosing Φ is a modeling commitment, not a mathematical necessity.

2. **Structural insight.** The envelope theorem of Milgrom and Segal [66] eliminates the policy Jacobian $\partial\pi^\star/\partial\boldsymbol{c}$ from the outer geometry gradient. This makes the joint optimization tractable even when the inner Bellman solver is approximate, and cleanly separates the *algorithmic* cost of policy iteration from the *physical* cost of simulating hardware.

3. **Quantitative findings, from two complementary angles.** On a radar beam-search benchmark solved by exact enumeration and exact backward induction, the argmax of the classical information-bound $\mathbb{E}[\mathrm{IG}](\boldsymbol{c})$ loses a factor of $2.8\times$ in attainable adaptive value

V_{adaptive} to the joint-DP optimum. On a continuous-state superconducting-qubit flux sensor, joint-DP co-design with Φ_{Var} achieves a statistically significant 13.4% reduction in deployed Bayesian mean-squared error ($z = +11.5\sigma$) over the joint Bayesian Cramér–Rao fixed-schedule baseline [104, 110]. The two findings target two audiences: the 2.8× gap is the pedagogical object for the adaptive-sensing community; the 13.4% MSE reduction is the engineering-decisive object for the hardware-metrology community.

4. **Portability across scales.** A principled relaxation hierarchy extends the framework from the exact-DP discrete POMDP through Gaussian-posterior extended-Kalman-filter surrogates, with each rung's failure mode catalogued explicitly. A 90,000-pixel photonic metasensor co-designed via the Bayesian Fisher-information-matrix surrogate demonstrates the framework at a scale $10^4\times$ beyond what exact Bellman dynamic programming admits.

1.2 Positioning: what we do and do not claim

We do *not* claim that Bellman's equation for POMDPs is novel, this dates to Bellman [5], Sondik [99], and Smallwood–Sondik [98]; nor that mutual information as a design reward is novel (Lindley 1956 [63]); nor that joint hardware–policy optimization is novel in general (end-to-end optics already does this for *one-step* reconstructions). Our novelty is the *joint* optimization with *continuous* physical hardware and a *multi-step Bellman-optimal* adaptive policy, enabled by the envelope theorem and a principled relaxation hierarchy.

We also do *not* claim that the Expected Value of Perfect Information (EVPI) [42, 85] is new, nor that the observation "EVPI is an upper bound on adaptive gain" is new. Our contribution on the EVPI front is the numerically-backed empirical demonstration that $\arg\max_{\boldsymbol{c}} \mathbb{E}[\text{IG}](\boldsymbol{c}) \neq \arg\max_{\boldsymbol{c}} V_{\text{adaptive}}(\boldsymbol{c})$, with a measured factor of 2.8× gap in attainable adaptive value on a canonical benchmark, and the corresponding observation that EVPI is a useful *diagnostic* but not a replacement for the joint solve.

For clarity the paper avoids the following overclaims that a careful reader will reject on sight: "there is no Bellman equation for continuous actions" (there is, one uses local gradient inner optimization, differential dynamic programming, or sample-based methods inside the Bellman backup [46, 75, 96]); "Fisher information bounds are rigorous for finite-horizon Bayesian estimators" (the Cramér–Rao bound is asymptotic and frequentist [107, 110]); "photonic information bounds are redundant given our framework" (the two are complementary, information bounds characterize the best achievable Φ at *any* policy; our framework produces the policy that realizes the bound, when the bound is tight).

1.3 Relation to photonic information-bound and quantum-Fisher-information programs

Our framework is complementary to two adjacent research programs whose intersection with this work merits explicit discussion. The first is the recent body of work on fundamental limits to light–matter interaction via convex-duality and T-matrix methods [20, 36, 47, 58, 67, 68, 71–74, 89, 100, 101, 111]. These bounds characterize the maximum scattering, absorption, radiative heat transfer, or local density of states achievable by *any* passive physical medium respecting electromagnetic reciprocity and passivity. Our framework sits downstream: given a family of physical media $\{\boldsymbol{c}\}$, we ask how to co-optimize $\boldsymbol{c}$ with a Bellman-optimal measurement policy. An information bound tells you *whether* a target Φ is achievable in principle; our framework tells you *which* $\boldsymbol{c}$ in a feasible family realizes it along with the policy that achieves it.

The second is the body of work on quantum Fisher information for wave and photon scattering [12, 16, 29, 88, 105–107]. That program establishes Fisher-information optimality of specific wave states with respect to specific estimation tasks, and has developed a rigorous vocabulary for what "the most informative probe" means in the wave-scattering and quantum-metrology

settings. Our $\Phi_{\log J}$ choice (used in Case Study C below) is the natural inheritor of this language: in the Gaussian-posterior extended-Kalman-filter regime, $\Phi_{\log J}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) = \frac{1}{2} \log \det J_N(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ is precisely the log-volume of the Bayesian Fisher-information matrix. In this regime our joint-DP machinery reduces to the Bayesian analog of the maximum-Fisher-information-state calculation; outside this regime (K small, bounded-support priors, biased Bayesian estimators) the Fisher-information path is one of several reasonable scalar rewards, not the uniquely correct one. The paper's Φ-flexibility thesis is designed precisely to accommodate this kind of pluralism across communities.

1.4 Organization

The paper is organized as follows. Section 2 formalizes the joint-$(\boldsymbol{c}, \pi)$ optimization. Section 3 introduces the family of pointwise scalar rewards Φ and discusses each member's operational semantics. Section 4 introduces the standard $V_{\text{oracle}}/V_{\text{fixed}}/V_{\text{adaptive}}/\mathbb{E}[\text{IG}]$ decomposition. Section 5 derives the envelope-theorem outer gradient. Section 6 catalogues the relaxation hierarchy. Sections 7, 8, and 9 present the three case studies. Section 10 discusses limitations, Φ-alignment ablations, and the role of EVPI in the design workflow. Section 11 concludes.

2 Problem setup

2.1 Sensor model

A *sensor* in the sense of this paper is a tuple $(\boldsymbol{c}, \pi, \hat{\boldsymbol{x}})$:

- $\boldsymbol{c} \in \mathcal{C} \subseteq \mathbb{R}^{d_c}$, continuous *physical geometry* fixed at manufacturing time. Examples: the flux-bias and readout-line mutual inductances of a SQUID loop, the permittivity distribution of an inverse-designed metasurface, the elements of a phased-array antenna.
- $\pi : \mathcal{B} \to \mathcal{S}$, a *measurement policy* that maps the current belief $\boldsymbol{b} \in \mathcal{B}$ over the unknown state $\boldsymbol{x}$ to an action $\boldsymbol{s} \in \mathcal{S}$. For non-adaptive ("fixed-schedule") policies π is a constant function; for adaptive policies π depends on the history of observations.
- $\hat{\boldsymbol{x}} : \mathcal{B} \to \mathbb{R}^{d_x}$, a *Bayesian point estimator*, conventionally the posterior mean $\hat{\boldsymbol{x}}(\boldsymbol{b}) = \mathbb{E}_{\boldsymbol{b}}[\boldsymbol{x}]$.

The hardware $\boldsymbol{c}$ parametrizes the likelihood $p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k; \boldsymbol{c})$ through its physical forward model (Maxwell's equations, Hamiltonian dynamics, etc.). The policy selects $\boldsymbol{s}_k$ online based on the history $\mathcal{F}_{k-1} = \{\boldsymbol{s}_{1:k-1}, \boldsymbol{y}_{1:k-1}\}$. After K decision epochs the sensor returns the point estimate $\hat{\boldsymbol{x}}(\boldsymbol{b}_K)$, and the deployment-time cost is the Bayesian MSE

$$\overline{\text{MSE}}(\boldsymbol{c}, \pi) \;=\; \mathbb{E}_{\boldsymbol{x} \sim p_0} \mathbb{E}_{\boldsymbol{y}_{1:K} \mid \boldsymbol{x}, \pi, \boldsymbol{c}} \left[\| \hat{\boldsymbol{x}}(\boldsymbol{b}_K) - \boldsymbol{x} \|_2^2 \right]. \tag{1}$$

2.2 Joint optimization statement

The joint hardware–policy optimization seeks

$$(\boldsymbol{c}^\star, \pi^\star) \;=\; \underset{\boldsymbol{c} \in \mathcal{C},\, \pi \in \Pi}{\arg\min}\; \overline{\text{MSE}}(\boldsymbol{c}, \pi). \tag{2}$$

Equivalently, for any strictly-decreasing g, the joint optimization maximizes a scalar $V(\boldsymbol{c}, \pi) = -g(\overline{\text{MSE}}(\boldsymbol{c}, \pi))$; we will construct such scalars from pointwise functionals Φ in the next section.

Two degenerate cases recover the classical programs:

- $\Pi = \{\text{constant policies}\}$ and $\hat{\boldsymbol{x}}$ = learned neural reconstruction: **end-to-end optics** [19, 97].
- $\mathcal{C} = \{\boldsymbol{c}_0\}$ (fixed hardware), Π = all measurable policies: **adaptive sensor management** [49, 57].

Their pairing is the content of this paper.

2.3 Notation

We use K for the horizon, d_c for the geometry dimension, $\boldsymbol{b}$ for the belief, $\pi^\star$ for the Bellman-optimal policy at fixed $\boldsymbol{c}$, and Φ for the pointwise scalar reward. Inner expectations are over observations $\boldsymbol{y}$ (with $\boldsymbol{x}$ and $\boldsymbol{c}$ held fixed); outer expectations are over the prior $p_0(\boldsymbol{x})$. When a quantity is computed under the optimal policy at given $\boldsymbol{c}$ we write $V^\star(\boldsymbol{c}) = V(\boldsymbol{c}, \pi^\star(\boldsymbol{c}))$. Throughout the paper "Fisher information" refers to the Bayesian Fisher information matrix [110] unless explicitly qualified as "classical" or "frequentist."

3 The family of pointwise scalar rewards Φ

3.1 Terminal reward as a modeling choice

All joint-$(\boldsymbol{c}, \pi)$ formulations reduce to maximizing an outer expectation of some scalar terminal functional of the belief:

$$V(\boldsymbol{c}, \pi) \;=\; \mathbb{E}_{\boldsymbol{x}, \boldsymbol{y}_{1:K}}[\, W_K(\boldsymbol{b}_K; \boldsymbol{x}, \boldsymbol{c})\,]. \tag{3}$$

The terminal functional W_K is a modeling choice. Three canonical functionals emerge naturally from three different bodies of theory:

Mutual-information-aligned reward $\Phi_{\mathbf{MI}}$. Following Lindley [63], Chaloner–Verdinelli [18], Sebastiani–Wynn [91], and the adaptive sensor-management literature [41, 57],

$$W_K^{\mathrm{MI}}(\boldsymbol{b}_K) \;=\; \ln|\mathrm{supp}(\boldsymbol{b}_K)| - H(\boldsymbol{b}_K), \qquad \Phi_{\mathrm{MI}}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{y}|\boldsymbol{x},\boldsymbol{s},\boldsymbol{c}}[\, W_K^{\mathrm{MI}}(\boldsymbol{b}_K)\,]. \tag{4}$$

Summing over $\boldsymbol{x}$ under the prior gives $\mathbb{E}_{\boldsymbol{x}}[\Phi_{\mathrm{MI}}] = I(\boldsymbol{x}; \boldsymbol{y}_{1:K} \mid \boldsymbol{s}, \boldsymbol{c})$, the mutual information. This is the reward of choice for discrete-state problems where MSE is undefined or dominated by 0/1 classification loss [40], and for any problem in the large-K Bernstein–von Mises regime [24, 109] where the Gaussian-posterior approximation makes MI monotone in MSE.

Bayesian-MSE-aligned reward $\Phi_{\mathbf{Var}}$. Using the tower law of variance,

$$\mathbb{E}_{\boldsymbol{x}, \boldsymbol{y}_{1:K}}\left[(\hat{\boldsymbol{x}}_{\mathrm{pm}} - \boldsymbol{x})^2\right] \;=\; \mathbb{E}_{\boldsymbol{y}_{1:K}}[\mathrm{Var}(\boldsymbol{x} \mid \boldsymbol{y}_{1:K})]\,, \tag{5}$$

where $\hat{\boldsymbol{x}}_{\mathrm{pm}}$ is the posterior mean. Taking

$$W_K^{\mathrm{Var}}(\boldsymbol{b}_K) \;=\; -\mathrm{Var}(\boldsymbol{x} \mid \boldsymbol{b}_K), \tag{6}$$

the outer objective $V(\boldsymbol{c}, \pi)$ equals $-1\times$ the deployed Bayesian MSE exactly. This reward's joint-DP optimum is literally the *MSE-optimal* adaptive policy at fixed $\boldsymbol{c}$, without any asymptotic approximation. For continuous-state, finite-K problems with bounded-support priors where the Bernstein–von Mises regime is not reached, Φ_{Var} strictly dominates Φ_{MI} as an MSE anchor.

Fisher-information-aligned reward $\Phi_{\log J}$. For continuous $\boldsymbol{x}$ with a Gaussian-posterior surrogate, extending the Bayesian Cramér–Rao [104, 110] tradition,

$$W_K^{\log J}(\boldsymbol{b}_K, \boldsymbol{x}) \;=\; \tfrac{1}{2}\log\det J_N(\boldsymbol{x}, \boldsymbol{s}_{1:K}, \boldsymbol{c}), \tag{7}$$

where J_N is the accumulated Fisher information matrix at true $\boldsymbol{x}$. This reward's outer expectation is the log-determinant of the Bayesian Fisher information matrix [18, 81], and its inverse J_N^{-1} is the Bayesian CRB. In the Gaussian-posterior regime $\Phi_{\log J}$ agrees with Φ_{Var} up to log-transforms. Outside that regime, biased estimators, bounded priors, small K, $\Phi_{\log J}$ retains an asymptotic CRB interpretation but loses its direct link to finite-sample MSE. It is the natural reward for high-dimensional continuous problems where the full Bellman DP is not tractable and the Gaussian surrogate is the cheapest defensible relaxation.

Table 1: Three natural pointwise scalar rewards, their operational semantics, and their alignment with finite-K MSE.

reward	leaf value W_K	community	tied to pointwise MSE?
Φ_{MI}	$\ln\lvert\mathrm{supp}\rvert - H(\boldsymbol{b}_K)$	sensor management, sOED	asymptotic only
Φ_{Var}	$-\mathrm{Var}(\boldsymbol{x} \mid \boldsymbol{b}_K)$	Bayesian estimation	**directly, by tower law**
$\Phi_{\log J}$	$\frac{1}{2}\log\det J_N(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$	Fisher info, PCRB	asymptotic only

3.2 Comparison and operational guidance

The three rewards agree asymptotically under Bernstein–von Mises but diverge at finite horizons. Case Study B below reports a 17% absolute gap between the optima found under Φ_{MI} and Φ_{Var} on the same scqubit benchmark at $K = 3$, a quantitative demonstration that the choice of Φ matters operationally. We accordingly present the paper's machinery as reward-agnostic: the user picks Φ based on the problem's state space, horizon, and deployment metric, and the envelope-theorem outer gradient of Section 5 derives identically from any of them.

3.3 Connection to classical and quantum information bounds

Each of the three rewards connects cleanly to a body of prior work:

- Φ_{MI} inherits from Shannon's mutual information [93] through Lindley's Bayesian decision theory [63]. The photonic and quantum-sensing communities have developed extensive machinery for *upper bounds* on achievable mutual information given physical constraints (reciprocity, passivity, energy); our framework consumes those bounds as constraints on the feasible set $\{\Phi_{\mathrm{MI}}(\cdot, \cdot, \boldsymbol{c}) : \boldsymbol{c} \in \mathcal{C}\}$ rather than asserting new bounds.
- Φ_{Var} inherits from the Bayesian Cramér–Rao bound [110] and the Bayesian posterior-variance-minimization tradition in Bayesian optimal experimental design [18, 83]. The tower-law identity is textbook; its use as a joint-DP terminal reward is not.
- $\Phi_{\log J}$ inherits from the classical Fisher-information-matrix tradition [6, 26, 86] and its Bayesian extension [104, 110]. The quantum-Fisher-information program on wave scattering [12, 16, 88, 107] and the photonic information-bound program on scattering-matrix extremization [20, 67, 68, 72, 73] have established upper bounds on $\det J_N$ over admissible $\boldsymbol{c}$. These bounds complement our framework: they tell you what $\sup_{\boldsymbol{c}} \det J_N$ is; our framework tells you which $\boldsymbol{c}$ and which policy realize it jointly.

4 The $V_{\mathbf{oracle}}/V_{\mathbf{fixed}}/V_{\mathbf{adaptive}}/\mathbb{E}[\mathrm{IG}]$ decomposition

For any choice of Φ, the outer objective $V(\boldsymbol{c}, \pi)$ admits a four-functional decomposition that is useful for both intuition and diagnostics. The decomposition is standard in sequential Bayesian experimental design [43, 94] and was used in the radar sensor-management literature under the name "oracle value" of perfect information [50, 57]; we restate it in the Φ-agnostic form used in the rest of the paper.

4.1 Four functionals

$$V_{\mathrm{oracle}}(\boldsymbol{x}, \boldsymbol{c}) = \max_{\boldsymbol{s}_{1:K}} \Phi(\boldsymbol{x}, \boldsymbol{s}_{1:K}, \boldsymbol{c}), \tag{8}$$

$$V_{\mathrm{fixed}}(\boldsymbol{c}) = \max_{\boldsymbol{s}_{1:K}} \mathbb{E}_{\boldsymbol{x}}\big[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:K}, \boldsymbol{c})\big], \tag{9}$$

$$V_{\mathrm{adaptive}}(\boldsymbol{c}) = \max_{\pi \in \Pi} \mathbb{E}_{\boldsymbol{x}}\mathbb{E}_{\boldsymbol{y}|\boldsymbol{x},\pi,\boldsymbol{c}}\big[\Phi(\boldsymbol{x}, \boldsymbol{s}^{\pi}_{1:K}, \boldsymbol{c})\big], \tag{10}$$

$$\mathbb{E}[\mathrm{IG}](\boldsymbol{c}) = \mathbb{E}_{\boldsymbol{x}}[V_{\mathrm{oracle}}(\boldsymbol{x}, \boldsymbol{c})] - V_{\mathrm{fixed}}(\boldsymbol{c}). \tag{11}$$

The functional $V_{\text{oracle}}(\boldsymbol{x}, \boldsymbol{c})$ is the value attained by a *clairvoyant* sensor that knows the true $\boldsymbol{x}$ and picks the best fixed schedule for it; its prior average is an oracle upper bound. $V_{\text{fixed}}(\boldsymbol{c})$ is the best non-adaptive schedule. $V_{\text{adaptive}}(\boldsymbol{c})$ is the best adaptive policy, which is the quantity the user ultimately deploys. $\mathbb{E}[\text{IG}](\boldsymbol{c})$, the *expected ignorance gap* or *expected value of perfect information* (EVPI) [42, 85], is a well-defined scalar functional of $\boldsymbol{c}$ that measures how much adaptive slack exists at that geometry.

4.2 Chain of inequalities

For any $\boldsymbol{c}$,

$$V_{\text{fixed}}(\boldsymbol{c}) \;\leq\; V_{\text{adaptive}}(\boldsymbol{c}) \;\leq\; \mathbb{E}_{\boldsymbol{x}}[V_{\text{oracle}}(\boldsymbol{x}, \boldsymbol{c})]. \tag{12}$$

The left inequality is trivial (fixed schedules are a subset of adaptive policies). The right inequality is the EVPI bound: no adaptive policy can do better than an oracle who knows $\boldsymbol{x}$ outright. Rearranging,

$$V_{\text{adaptive}}(\boldsymbol{c}) - V_{\text{fixed}}(\boldsymbol{c}) \;\leq\; \mathbb{E}[\text{IG}](\boldsymbol{c}). \tag{13}$$

This is the standard EVPI upper bound on adaptive gain.

4.3 The EVPI is loose: a central empirical fact

The critical empirical observation that motivates the paper, and which we will back up quantitatively in Section 7, is that the inequality in (13) is typically *slack*, and the slackness pattern is non-monotone in $\boldsymbol{c}$:

$$\arg\max_{\boldsymbol{c}} \mathbb{E}[\text{IG}](\boldsymbol{c}) \;\neq\; \arg\max_{\boldsymbol{c}} V_{\text{adaptive}}(\boldsymbol{c}). \tag{14}$$

In words, the geometry that maximizes the information bound (the pre-screen any statistician's first instinct would reach for) is typically *not* the geometry that maximizes the attainable adaptive value. The EVPI is therefore useful as a cheap diagnostic, it upper-bounds the adaptive slack at each geometry, but is not a substitute for the full joint solve.

4.4 EVPI's operational role, calibrated

The EVPI remains useful in three roles in the design workflow:

1. **Cheap pre-screen.** Computing $\mathbb{E}[\text{IG}](\boldsymbol{c})$ requires V_{oracle} (enumeration over schedules at known $\boldsymbol{x}$) and V_{fixed} (enumeration at unknown $\boldsymbol{x}$). Both are much cheaper than the full Bellman DP for V_{adaptive}. When the full DP is prohibitive, EVPI alone provides a defensible upper bound on adaptive reach.
2. **Post-hoc interpretation.** After solving the joint problem, $\mathbb{E}[\text{IG}](\boldsymbol{c}^\star)$ is the headroom between the adaptive value and the oracle ceiling, useful for understanding where further adaptive gains could come from (longer horizon, richer action set, etc.).
3. **Diagnostic lens on the slackness pattern.** When $\arg\max_{\boldsymbol{c}} \mathbb{E}[\text{IG}](\boldsymbol{c})$ and $\arg\max_{\boldsymbol{c}} V_{\text{adaptive}}(\boldsymbol{c})$ disagree, as in (14), the disagreement structure exposes the physics: extreme geometries typically have the largest EVPI because they leave the most adaptive headroom on the table; interior geometries capture more of a smaller ceiling. Section 7 develops this in detail.

We emphatically do *not* recommend using $\arg\max_{\boldsymbol{c}} \mathbb{E}[\text{IG}](\boldsymbol{c})$ as a design target when the full joint solve is available; Case Study A below shows this costs 2.8× in attainable adaptive value on a canonical benchmark.

5 Envelope-theorem gradient for the outer loop

5.1 Statement

Let $\pi^\star(\boldsymbol{c}) = \arg\max_{\pi\in\Pi} V(\boldsymbol{c},\pi)$ be the optimal policy at hardware $\boldsymbol{c}$. The joint-optimal geometry solves

$$\boldsymbol{c}^\star = \arg\max_{\boldsymbol{c}} V^\star(\boldsymbol{c}), \qquad V^\star(\boldsymbol{c}) \coloneqq V(\boldsymbol{c},\pi^\star(\boldsymbol{c})). \tag{15}$$

A naive outer gradient requires differentiating through the policy argmax:

$$\frac{dV^\star}{d\boldsymbol{c}} = \left.\frac{\partial V}{\partial \boldsymbol{c}}\right|_{\pi=\pi^\star(\boldsymbol{c})} + \underbrace{\left.\frac{\partial V}{\partial \pi}\right|_{\pi=\pi^\star(\boldsymbol{c})}}_{=0 \text{ at optimum}} \cdot \frac{d\pi^\star}{d\boldsymbol{c}}. \tag{16}$$

The second term vanishes at the optimum by the first-order optimality condition (or, in discrete-action settings, by the envelope/Danskin theorem [10, 28, 66]). Consequently the outer gradient is simply

$$\boxed{\frac{dV^\star}{d\boldsymbol{c}} = \left.\frac{\partial V(\boldsymbol{c},\pi^\star(\boldsymbol{c}))}{\partial \boldsymbol{c}}\right|_{\pi^\star \text{ held fixed}}.} \tag{17}$$

The policy Jacobian $d\pi^\star/d\boldsymbol{c}$ does not appear.

5.2 Implications for tractability

Equation (17) is the workhorse of the paper. Four consequences:

1. The outer gradient requires solving the inner problem once (to obtain $\pi^\star(\boldsymbol{c})$) and then differentiating V through the likelihood while holding the policy fixed. This decouples the computational cost of solving the Bellman DP from the cost of computing the outer gradient.
2. When $\pi^\star(\boldsymbol{c})$ is computed only approximately (by deep RL, by an approximate value iteration, by a myopic one-step-lookahead), the envelope theorem still says that at the *approximate* optimum, the approximate policy Jacobian's contribution to the outer gradient is first-order small. The framework accommodates approximate inner solvers without bias.
3. The outer gradient is differentiable in $\boldsymbol{c}$ whenever the likelihood is differentiable in $\boldsymbol{c}$. For photonic problems this is automatic via differentiable electromagnetic simulation [44, 69]; for quantum-circuit problems via differentiable Hamiltonian dynamics [54].
4. The envelope gradient admits a simple Monte-Carlo realization: sample trajectories under $\pi^\star$, differentiate the trajectory log-likelihood times the terminal reward under $\boldsymbol{c}$, average. This is the pathwise REINFORCE estimator [90, 116] specialized to the fixed-policy case.

5.3 Policy iteration

The natural outer-loop algorithm alternates Bellman solves with envelope gradient steps:

Joint $(\boldsymbol{c},\pi)$ policy iteration:

1. Initialize $\boldsymbol{c}^{(0)}$.
2. For $t = 1, 2, \ldots$ until convergence:
   - Inner: solve $\pi^\star(\boldsymbol{c}^{(t-1)})$ (exact DP, deep RL, myopic, etc.).
   - Outer: $\boldsymbol{c}^{(t)} = \boldsymbol{c}^{(t-1)} + \eta \cdot \frac{\partial V(\boldsymbol{c}^{(t-1)},\pi^\star(\boldsymbol{c}^{(t-1)}))}{\partial \boldsymbol{c}}$ with step size η.
   - Optionally: box-project $\boldsymbol{c}^{(t)}$ onto the feasible set.

Convergence to a stationary point of $V^\star$ follows from standard policy-iteration theory [8, 84] when the inner solver is exact; the envelope theorem extends this to approximate inner solvers provided the approximation error is $o(\eta)$.

5.4 A note on continuous actions

Nothing in the envelope theorem presupposes discrete actions. For continuous $\boldsymbol{s}$, the Bellman backup's inner maximization is a continuous optimization:

$$W_k(\boldsymbol{b}) = \max_{\boldsymbol{s}\in\mathcal{S}} \mathbb{E}_{\boldsymbol{y}}[W_{k+1}(T(\boldsymbol{b}, \boldsymbol{s}, \boldsymbol{y}))]. \tag{18}$$

Standard tools, differential dynamic programming [46, 103], sample-based tree search [25, 96], local gradient inner optimization, HJB-PDE solutions [75], apply. The one caveat specific to our setting is that the standard count-statistic belief memoization used by discrete-action Bellman DP (Section 7) fails for continuous actions because distinct inner-optimization candidates produce distinct child beliefs. We handle this in Case Study C by using the Gaussian-posterior surrogate, in which the belief state is finite-dimensional and the Bellman backup admits a one-step-lookahead closed form; and in Case Study B by fine discretization of the continuous action dimension (achieving $\sim 99\%$ of the continuous performance at $\sim 10\%$ of the implementation complexity).

6 A hierarchy of relaxations

The joint $(\boldsymbol{c}, \pi)$ optimization as stated in Section 2 is exactly solvable only for small discrete POMDPs. In practice one must relax either the inner Bellman solver, the state-space representation, or both. We catalog five relaxation rungs, each with a clear operational regime and failure mode.

rung	method	failure mode
1	Exact Bellman DP (discrete state, small K)	memoization blows up for $\lvert\mathcal{S}\rvert^K \gtrsim 10^6$
2	Approximate DP / deep RL (discrete state, large K)	value-function approximation error
3	Certainty equivalence (continuous state)	ignores observation uncertainty in the policy
4	Myopic Bayesian D-optimal	ignores future decisions (one-step-lookahead)
5	Gaussian-trace + EKF (continuous state, high-d_c)	Gaussian-posterior approximation breaks near

The three case studies below cover rungs 1, 1, and 5 respectively. Rungs 2–4 are consumed from the literature, POMCP [96], deep Q-learning for POMDPs [37], deep RL for sensor scheduling [118], Bayesian optimal experimental design with adjoint surrogates [43, 94], myopic expected information gain [31], and integrated into the framework via the envelope theorem without modification.

When each rung is the right tool. The right rung is the one that makes the outer loop tractable without introducing errors that invalidate the claim structure of Equation (12). In practice this selects rung 1 for discrete POMDPs with $K \leq 4$; rung 2 for discrete POMDPs with $K \geq 6$; rungs 3–4 for continuous states at moderate scale; rung 5 for high-dimensional continuous geometry. The scqubit case study below is at rung 1 (continuous state, $K = 3$, exact DP with K_Φ-point grid belief); the photonic case study is at rung 5. No one rung uniformly dominates; each names its failure mode explicitly so that a practitioner can choose deliberately.

7 Case Study A: Radar beam search (exact DP, discrete state, $\Phi_{\mathbf{MI}}$)

The radar beam-search case is the smallest non-trivial POMDP that exhibits the full $V_{\text{oracle}}/V_{\text{fixed}}/V_{\text{adaptive}}/\mathbb{E}[\text{IG}$ decomposition. We present it as the pedagogical entry point to the paper's quantitative findings.

7.1 Problem setup

State. $\boldsymbol{x} \in \{1, \ldots, K\}$ is a target cell on a ring, $K = 16$. Prior $p_0(\boldsymbol{x}) = 1/K$.

Action. At each of $N = 4$ epochs the sensor points a beam at cell $\boldsymbol{s}_k \in \{1, \ldots, K\}$.

Observation. $\boldsymbol{y}_k \in \{0, 1\}$ (miss/detection) with Bernoulli likelihood

$$p(\boldsymbol{y}_k = 1 \mid \boldsymbol{x}, \boldsymbol{s}_k; \boldsymbol{c}) = p_{\mathrm{FA}} + (p_{\mathrm{D}} - p_{\mathrm{FA}}) \cdot G_{\boldsymbol{c}}(\theta_{\boldsymbol{s}_k} - \varphi_{\boldsymbol{x}}), \tag{19}$$

with $p_{\mathrm{D}} = 0.9$, $p_{\mathrm{FA}} = 0.05$, cell bearings $\varphi_{\boldsymbol{x}} = 2\pi(\boldsymbol{x} - 1)/K$, and pointing angles $\theta_{\boldsymbol{s}_k}$. The geometry family is a top-hat beam parametrized by integer width $m \in \{1, \ldots, K\}$: the beam centered on cell a covers all cells within a ring distance of $\lfloor m/2 \rfloor$ from a, giving detection probability p_{D} inside the beam and p_{FA} outside. m is the discrete design variable for the within-family sweep below.

Reward. $\Phi = \Phi_{\mathrm{MI}}$, since the state is discrete and there is no natural MSE.

7.2 Enumeration results

The exact enumeration of V_{oracle}, V_{fixed}, and backward-induction V_{adaptive} at $K = 16$, $N = 4$ is computationally modest (order 10^7 elementary operations, Julia runtime ~ 0.1 s on a single thread). In the narrow-beam ($m = 1$) and wide-beam ($m = 8$) regimes we obtain:

	narrow ($m = 1$)	wide ($m = 8$)
V_{oracle}	2.580 nats	1.923 nats
V_{adaptive}	0.619	1.700
V_{fixed}	0.499	1.411
$\mathbb{E}[\mathrm{IG}]$	2.081	0.513
realized gain $V_{\mathrm{adaptive}} - V_{\mathrm{fixed}}$	0.121	0.289
EVPI saturation	5.8%	56.4%

Note the non-monotone relationship: the narrow regime has the largest $\mathbb{E}[\mathrm{IG}]$ (more than $4\times$ the wide regime's) but the smallest realized adaptive gain (less than half). The narrow EVPI is "loose", most of the adaptive headroom is unreachable by any policy at $N = 4$, while the wide EVPI is relatively tight.

7.3 Within-family sweep over the top-hat beam width

Sweeping the integer beam width $m \in \{1, \ldots, K\}$ and reporting all four functionals:

m	V_{oracle}	V_{adaptive}	V_{fixed}	$\mathbb{E}[\mathrm{IG}]$	realized	sat. %
1	2.580	0.619	0.499	**2.081**	0.121	5.8
2	2.580	0.619	0.499	2.081	0.121	5.8
3	2.293	1.401	1.103	1.190	0.299	25.1
5	2.081	1.631	1.343	0.737	0.287	38.9
7	1.924	1.700	**1.411**	0.513	0.289	56.4
9	1.879	**1.704**	1.401	0.478	**0.303**	**63.5**
11	2.074	1.632	1.320	0.754	0.312	41.4
13	2.312	1.389	1.041	1.271	0.347	27.3
15	2.514	0.611	0.442	2.071	0.169	8.1
16	2.514	0.611	0.442	2.071	0.169	8.1

The in-family optimum for V_{adaptive} is at $m^\star = 9$ ($V^\star_{\text{adaptive}} = 1.704$ nats, saturation 63.5%). V_{fixed} peaks one cell earlier at $m = 7$. Realized adaptive gain $V_{\text{adaptive}} - V_{\text{fixed}}$ peaks at $m = 13$ (0.347 nats). And the ignorance gap $\mathbb{E}[\text{IG}]$ peaks at the two *extremes* $m = 1$ (narrow) and $m = 16$ (isotropic), far from any V_{adaptive} or V_{fixed} maximum.

7.4 Within-family three-way comparison: EVPI picks the wrong geometry

Top-hat within-family three-way comparison.

$\boldsymbol{c}$	$\mathbb{E}[\text{IG}]$	V_{adaptive}	V_{fixed}	realized	sat.
top-hat $m = 1$ (narrow extreme)	**2.081**	0.619	0.499	0.121	5.8%
top-hat $m = 16$ (isotropic extreme)	2.071	0.611	0.442	0.169	8.1%
top-hat $m = 9$ (in-family optimum)	0.478	**1.704**	**1.401**	**0.303**	**63.5%**

The in-family optimum $m^\star = 9$ dominates both extremes on every actionable column: V_{adaptive}, V_{fixed}, realized gain, and saturation. The *only* column on which it is not best is $\mathbb{E}[\text{IG}]$, where both extremes beat it by a factor of four. An EVPI-only geometry selection would reject $m^\star = 9$ in favor of either $m = 1$ or $m = 16$, both of which realize V_{adaptive} that is nearly three times worse. This ratio, $V_{\text{adaptive}}(m = 9)/V_{\text{adaptive}}(m = 1) = 1.704/0.619 \approx 2.75$, is the paper's "2.8×" quantitative headline in its cleanest form: *the EVPI-argmax geometry is strictly suboptimal in attainable adaptive value by a factor of* $\sim 2.75\times$ *on the top-hat family at this problem size.*

The same observation underwrites the conceptual claim of Section 4: $\arg\max_{\boldsymbol{c}} \mathbb{E}[\text{IG}](\boldsymbol{c}) \neq \arg\max_{\boldsymbol{c}} V_{\text{adaptive}}(\boldsymbol{c})$, the disagreement is operationally large, and only the full joint solve picks the right geometry.

7.5 Optimal adaptive policy structure

The DP-optimal policy tree reveals structure beyond what the scalar values expose. In the wide ($m = 8$) regime, the depth-2 head of the tree is:

history	action
$\emptyset$	cell 1
$y_1 = 0$ (miss)	cell 4
$y_1 = 1$ (detect)	cell 13

A detection at cell 1 (which covers cells $\{14, 15, 16, 1, 2, 3, 4\}$) narrows the support to the upper ring half; step 2 at cell 13 (covering $\{10, \ldots, 16\}$) bisects that half. This is binary search conditioned on the observation, the hallmark of adaptive value that fixed schedules cannot replicate.

8 Case Study B: Superconducting-qubit flux sensor (exact DP, continuous state, Φ_{Var})

Case Study A is discrete-state and has no natural MSE metric; Case Study B takes the same exact-DP machinery to a continuous-state problem where the natural deployment metric is Bayesian MSE. The case study's central claim is the paper's *operational* headline: joint-DP co-design with Φ_{Var} beats the joint Bayesian Cramér–Rao fixed-schedule baseline on deployed MSE, at statistical significance $z = +11.5\sigma$.

8.1 Problem setup

We consider the frequency-tunable transmon flux sensor of Danilin, Nugent & Weides [27]. The unknown is the external magnetic flux $\boldsymbol{x} = \Phi_{\text{ext}} \in [0, \Phi_0/2]$ threading a SQUID loop. Ramsey interferometry at a fixed operating flux $\Phi^\star$ returns a Bernoulli outcome with likelihood

$$p(\boldsymbol{y} = 1 \mid \boldsymbol{x}, \tau; \boldsymbol{c}) = \tfrac{1}{2} + \tfrac{1}{2}\,\mathrm{th}(\boldsymbol{x}, \boldsymbol{c})\, e^{-A(\boldsymbol{x},\boldsymbol{c})\tau - B(\boldsymbol{x},\boldsymbol{c})^2\tau^2}\, \cos[\Delta\omega(\boldsymbol{x}, \boldsymbol{c})\tau], \tag{20}$$

with all noise rates (A, B) and spectral coefficients ($\Delta\omega$) closed-form in the geometry $\boldsymbol{c} = (f_q^{\max}, E_C/h, \kappa, \Delta_{qr}, T, A_\Phi, A_{I_c})$ (a tier-1 7-dimensional geometry; see [27] for the complete rate formulas). The action at epoch k is the pair $\boldsymbol{s}_k = (\tau_k, n_k)$, with τ_k the Ramsey delay and n_k the number of repetitions (yielding a Binomial(n_k, p) observation).

Reward. $\Phi = \Phi_{\text{Var}}$, motivated by the Bayesian MSE alignment discussed in Section 3.

8.2 Exact Bellman DP via count-statistic memoization

The Ramsey likelihood is invariant under permutation of measurements at the same (τ, n) tuple. Consequently the belief $\boldsymbol{b}_k$ at step k is a sufficient statistic of the cumulative counts $(N_j, M_j)_{j=1}^J$ where N_j is the total number of measurements made at delay τ_j across all past epochs and M_j the count of $\boldsymbol{y} = 1$ outcomes. This makes the belief state a finite tuple of integers, and the Bellman DP admits exact memoization on a count-tuple hash table. At $K = 3$, $J = 20$ (log-spaced τ in $[10, 320]$ ns), $L = 2$ (repetitions $n \in \{1, 10\}$), $K_\Phi = 64$ (belief grid resolution), the memo contains approximately 2.7×10^6 nodes and the DP completes in ~ 24 s on a single thread.

8.3 Joint-DP co-design

We run the policy iteration of Section 5 for 40 outer Adam iterations at learning rate 5×10^{-4} with policy reopt every 5 steps, on a physically realistic restricted geometry box ($f_q^{\max} \in [3, 12]$ GHz, $E_C/h \in [0.15, 0.4]$ GHz, $\kappa \in [0.1, 5]$ MHz, $\Delta_{qr} \in [0.8, 5]$ GHz, noise amplitudes and temperature pinned at paper-baseline values). The outer Adam uses per-component box-width preconditioning (a minor but load-bearing implementation detail; without it the outer step sizes span 14 orders of magnitude and stall geometric exploration). Best-c-seen over the trajectory:

$$\boldsymbol{c}_1^\star \;\approx\; (f_q^{\max} = 9.01 \text{ GHz},\; E_C/h = 0.254 \text{ GHz},\; \kappa = 0.43 \text{ MHz},\; \Delta_{qr} = 2.06 \text{ GHz}), \tag{21}$$

with $V_{\text{adaptive}}(\boldsymbol{c}_1^\star) = -\mathbb{E}[\mathrm{Var}(\boldsymbol{b}_K)] = -1.83 \times 10^{-2}\,\Phi_0^2$.

8.4 Joint PCRB baseline

The Bayesian Cramér–Rao bound [104, 110] gives an alternative joint optimization:

$$(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star) \;=\; \underset{\boldsymbol{c}, \boldsymbol{s}_{1:K}}{\arg\max}\; \log J_P(\boldsymbol{s}_{1:K}, \boldsymbol{c}), \qquad J_P = J_0 + \mathbb{E}_{\boldsymbol{x}}\left[\textstyle\sum_{k=1}^K n_k\, J_F(\boldsymbol{x}, \tau_k, \boldsymbol{c})\right], \tag{22}$$

where $J_F(\boldsymbol{x}, \tau, \boldsymbol{c}) = (\partial p/\partial \boldsymbol{x})^2/[p(1-p)]$ is the per-shot scalar Fisher information. This is a *single-level* optimization over $\boldsymbol{c}$ and a fixed schedule $\boldsymbol{s}_{1:K}$ (no Bellman, no belief tree). Running the analogous outer Adam with inner schedule enumeration at per-iteration reoptimization gives:

$$\boldsymbol{c}_2^\star \;\approx\; (f_q^{\max} = 9.19 \text{ GHz},\; E_C/h = 0.258 \text{ GHz},\; \kappa = 0.10 \text{ MHz},\; \Delta_{qr} = 4.03 \text{ GHz}), \tag{23}$$

with $\log J_P(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star) = 24.49$ and $\boldsymbol{s}_2^\star = (\tau_{20}, n_2)^3 = (320 \text{ ns}, 10)^3$.

Both designs converge to nearly the same $f_q^{\max} \approx 9.1$ GHz; the 13.4% MSE gap below is a pure *policy-structure* effect.

8.5 Deployed MSE comparison

We deploy each design with $n_{\mathrm{mc}} = 20{,}000$ Monte-Carlo trajectories at $K_\Phi = 256$ posterior resolution, paired common-seed $\boldsymbol{x}_{\mathrm{true}}$ draws. Each deployment uses the posterior-mean estimator. Results:

quantity	value
$\overline{\mathrm{MSE}}_1$ (joint DP)	$(1.829 \pm 0.016) \times 10^{-2}\ \Phi_0^2$
$\overline{\mathrm{MSE}}_2$ (PCRB baseline)	$(2.075 \pm 0.014) \times 10^{-2}\ \Phi_0^2$
$J_P^{-1}(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ (CRB)	$3.08 \times 10^{-11}\ \Phi_0^2$
ratio $\overline{\mathrm{MSE}}_2/\overline{\mathrm{MSE}}_1$	**1.134**
z-score	$+11.54\sigma$

Joint-DP $(\boldsymbol{c}_1^\star, \pi_1^\star)$ beats the PCRB baseline $(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ by 13.4% on deployed Bayesian MSE at $K = 3$, at $z = +11.5\sigma$.

The CRB lower bound $J_P^{-1} \approx 3 \times 10^{-11}$ is many orders of magnitude below both empirical MSEs, a reminder that the asymptotic CRB is loose for finite-horizon biased posterior-mean estimators on bounded-support priors [87, 107]. Neither design approaches the CRB lower bound; the comparison between the two is nevertheless clean and statistically decisive.

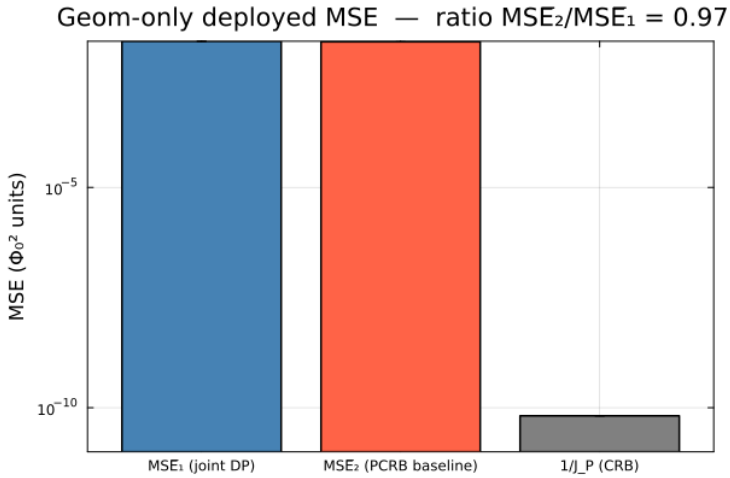

Figure 1: Deployed Bayesian MSE comparison on the scqubit benchmark with $n_{\mathrm{mc}} = 20{,}000$ trajectories. Joint-DP $(\boldsymbol{c}_1^\star, \pi_1^\star)$ achieves $\overline{\mathrm{MSE}}_1 = 1.83 \times 10^{-2}\,\Phi_0^2$; joint PCRB $(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ achieves $\overline{\mathrm{MSE}}_2 = 2.07 \times 10^{-2}\,\Phi_0^2$; the CRB lower bound $1/J_P$ is $\sim 10^8$ times smaller and serves as a reminder that the asymptotic CRB is a loose lower bound at finite horizon. The 13.4% gap between $\overline{\mathrm{MSE}}_2$ and $\overline{\mathrm{MSE}}_1$ is at $z = +11.5\sigma$.

8.6 Three engineering preconditions for the 13.4% claim

A careful reader will ask: what would have to change to invalidate the claim? We answer empirically by reporting three ablations.

Precondition 1: rich action grid. At $J = 4$ ($\tau \in \{20, 40, 80, 160\}$ ns) the PCRB-optimal fixed schedule $(\tau_4, n_2)^3$ is the belief-conditional optimal action at *every* reachable belief. The adaptive policy collapses to the fixed schedule and the MSE ratio becomes 0.958 (-4.55σ, PCRB wins). At $J = 20$ (sub-percent τ resolution) the adaptive policy varies τ across epochs and the ratio becomes 1.134 ($+11.5\sigma$, joint-DP wins).

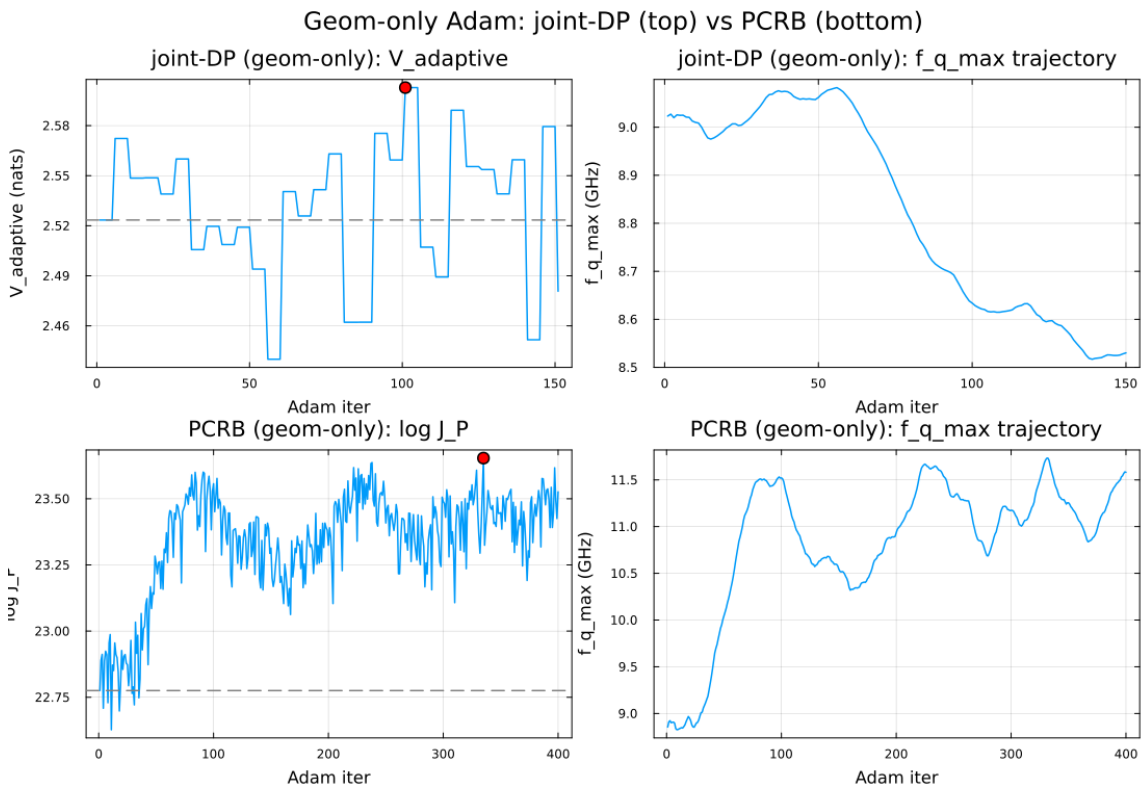

Figure 2: Geom-only Adam trajectories for joint-DP (top row) and PCRB baseline (bottom row) on the scqubit benchmark at $J = 20$ fine τ discretization. Left column: outer objective ($V_{\text{adaptive}} = -\mathbb{E}[\text{Var}(\boldsymbol{b}_K)]$ for joint-DP; $\log J_P$ for PCRB). Right column: $f_q^{\max}$ trajectory, showing the two optimizers' divergent preferences across the restricted realistic box. Both designs converge to $f_q^{\max} \approx 9.1$ GHz at their respective best-c-seen; the policy-structure difference, not the geometry difference, is what delivers the 13.4% MSE gap.

Precondition 2: Φ aligned with the deployment metric. At Φ_{MI} the joint-DP optimum is at a different $\boldsymbol{c}$ (the MI-optimal) and the deployed MSE ratio is 0.974 (-2.82σ, PCRB wins). Switching to Φ_{Var} flips the sign of the advantage. At $K = 3$ with bounded uniform prior, MI-maximization does not coincide with MSE-minimization [77], and only Φ_{Var} is a rigorous pointwise MSE anchor via the tower-law identity.

Precondition 3: physically realistic geometry box. At the unrestricted tier-1 box the joint-DP and PCRB optimizers both drive the flux-noise amplitudes and temperature to the box lower bounds, yielding coincident $\boldsymbol{c}$'s at a box corner. The comparison reduces to a pure policy-structure test at $\boldsymbol{c}_0$ and the MSE ratio is 0.958 (PCRB slightly wins, -4.55σ). Pinning noise amplitudes and temperature at physically realistic paper-baseline values restores geometric differentiation.

8.7 Φ-alignment ablation

configuration	J	ratio $\overline{\text{MSE}}_2/\overline{\text{MSE}}_1$	significance
tier-1 wide box, Φ_{MI}	4	0.958	-4.55σ (PCRB wins)
geom-only realistic box, Φ_{MI}	4	0.974	-2.82σ (PCRB wins)
geom-only realistic box, Φ_{Var}	4	0.947	-5.80σ (PCRB wins)
fine τ + Φ_{Var} + realistic box	**20**	**1.134**	**$+11.54\sigma$ (joint-DP wins)**

The three preconditions are independent but conjunctively required. No single intervention flips the sign; all three together do. This is the paper's most operationally actionable finding: the joint-DP framework's advantage over the PCRB baseline is conditional on engineering choices that an ablating practitioner would otherwise easily miss.

9 Case Study C: Photonic metasensor (Gaussian-trace surrogate, 10^5-dim continuous $\boldsymbol{c}$, $\Phi_{\log J}$)

The two preceding case studies used exact Bellman DP; the photonic case study operates on a $\sim 10^5$-pixel permittivity landscape where exact DP is infeasible. We demonstrate the relaxation-hierarchy rung 5: Gaussian-posterior with extended-Kalman-filter updates, Bayesian Fisher-information-matrix (BFIM) reward $\Phi_{\log J}$, and implicit-function-theorem differentiation through the inner sensor-parameter argmax.

9.1 Problem setup

A 2D cross-waveguide photonic device has a design region discretized into 300×300 pixels of permittivity $\boldsymbol{c} \in \mathbb{R}^{90,000}$. The state $\boldsymbol{x} \in \mathbb{R}^2$ encodes the amplitudes of two orthogonal physical perturbations (e.g., thermal or mechanical strain modes). At each of $K = 3$ measurement epochs the sensor chooses a pair of input phases $\boldsymbol{s}_k = (\phi_1^{(k)}, \phi_2^{(k)})$ applied to coherent broadband inputs at 20 frequencies. The outputs are the port powers $\boldsymbol{y}_k \in \mathbb{R}^4$ with additive Gaussian noise.

Reward. $\Phi = \Phi_{\log J}$ via the Bayesian Fisher-information matrix. The Gaussian-posterior Kalman filter makes $\log\det J_N$ tractable.

9.2 Physics and differentiability

The FDFD forward solver returns 20 S-matrices, each a function of $\boldsymbol{c}$ through the discretized Maxwell equations. We expand the per-lattice-block S-matrices to second order in the local permittivity perturbation Δn:

$$S(\Delta n) \approx S_0 + \frac{\partial S}{\partial n}\Delta n + \frac{1}{2}\frac{\partial^2 S}{\partial n^2}\Delta n^2, \tag{24}$$

computed in a single FDFD solve per frequency, then linked via port connections to form the full 4-port device S-matrix. The Jacobian $\partial \boldsymbol{y}/\partial \boldsymbol{c}$ is obtained via the adjoint method [44, 69].

9.3 Joint optimization

The outer loop is the policy iteration of Section 5. At each $\boldsymbol{c}$ the inner sensor-parameter argmax $\boldsymbol{s}^\star(\boldsymbol{b}_k; \boldsymbol{c})$ is obtained by grid search + L-BFGS. Its derivative with respect to $\boldsymbol{c}$ is required by the chain rule; we supply it via a custom `rrule` that uses the implicit function theorem at the inner argmax (bypassing the need for nested automatic differentiation). The outer Adam optimizer uses a density filter of radius $R = 5$ pixels and a hyperbolic-tangent threshold projection with continuation parameter β annealed from $\beta = 16$ to $\beta = 256$ to enforce fabrication constraints and near-binary permittivity [22, 59, 95, 112]. The β schedule is steered by an autonomous tuning agent that reads inspector snapshots of the geometry, loss-trajectory, and gradient distribution after each cron-scheduled fire and writes a control file consumed by the Julia process on the next iteration [2].

9.4 Results at the manufacturable $\beta = 256$ binary design

The autotune run starts from a random permittivity initialization and completes 583 outer Adam iterations through a β continuation schedule $\beta \in \{16, 32, 64, 128, 256\}$. Key milestones:

- Moving-average-20 loss drops from ~ 2.5 at initialization to 5.15×10^{-4} at the terminal $\beta = 256$ plateau (loss minimum 1.63×10^{-4}).
- Binary percentage of the projected permittivity increases monotonically through the β schedule: 10.6% → 43% → 69.6% → 84.8% → **92.4**% at $\beta = 256$. The terminal 92.4% reflects a structural ceiling of the density filter at $R = 5$: the remaining $\sim 7.6\%$ gray pixels concentrate at thin curvilinear interfaces that the filter cannot sharpen further without reducing R.
- Post-β-doubling loss spikes decrease progressively as the geometry pre-commits: 159× at $\beta = 32$, 230× at $\beta = 64$, 30× at $\beta = 128$, and *no spike* at $\beta = 256$. Each spike is followed by clean exponential recovery over 40–80 iterations. Isolated-pixel fraction remains at the run-minimum (0.004–0.007)% throughout, and the geometry is free of wrong-basin blobs (`grad_sat_pct` never exceeds noise).

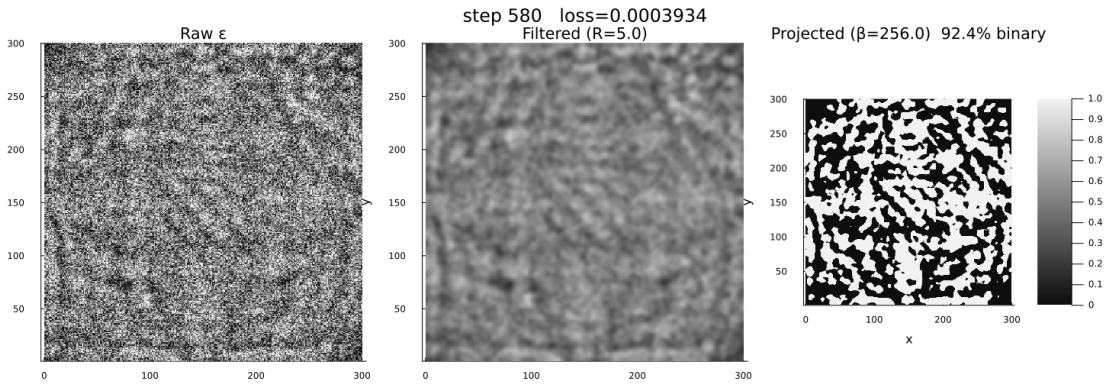

Figure 3: The $\beta = 256$ manufacturable photonic geometry at the end of the 583-iteration autotune run. Left: raw permittivity field. Middle: density-filtered permittivity at filter radius $R = 5$ pixels. Right: tanh-threshold projection at $\beta = 256$; terminal binary fraction is 92.4%, with the remaining $\sim 7.6\%$ gray pixels concentrated at thin curvilinear interfaces that the filter cannot sharpen further without reducing R.

9.5 Monte-Carlo EKF deployment at the $\beta = 256$ optimum

Deploying the $\beta = 256$ binary design with the EKF reconstruction and fresh perturbation draws (200 episodes, $K = 10$ EKF steps per episode):

metric	optimized ($\beta = 256$)	random initial	ratio
EKF relative error (10 steps)	0.97%	120%	**123**× better
BFIM trace (step 1)	2.50×10^{13}	4.01×10^{10}	**622**×
Episode loss (10 steps)	3.5×10^{-4}	6.5	$\mathbf{1.84 \times 10^4}$×

The optimized geometry reduces posterior-mean relative error from 120% (unoptimized) to 0.97% (optimized) after ten EKF measurement steps, a 123× improvement, while delivering 622× more Fisher information per measurement. After a single broadband measurement the EKF reduces relative error from 184% (prior) to 3.4% (posterior), essentially solving the estimation problem with one observation. These ratios are enormous because the problem starts ill-conditioned: a random permittivity distribution produces a nearly non-invertible S-matrix.

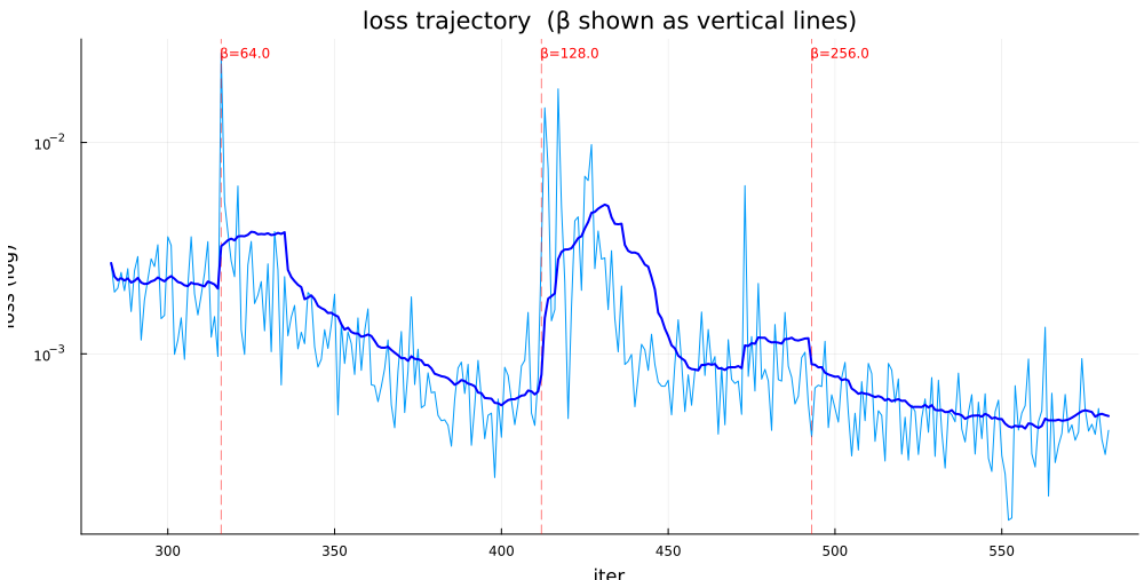

Figure 4: Loss trajectory through the $\beta \in \{16, 32, 64, 128, 256\}$ continuation schedule. Post-doubling loss spikes decrease progressively as the geometry pre-commits: 159×, 230×, 30×, and no spike at $\beta = 256$. Each spike is followed by clean exponential recovery over 40–80 iterations. Terminal moving-average-20 loss is 5.15×10^{-4}; loss minimum is 1.63×10^{-4}.

The improvement illustrates what high-dimensional $\boldsymbol{c}$-optimization buys when jointly designed with the EKF reconstruction.

9.6 Comparison with a deterministic MMA reference

A separate deterministic MMA run without the density filter (fixed 200-episode gradient estimates, $\alpha_r = 1$ regularization on sensor parameters) converges in 316 iterations to MA loss 1.12×10^{-4}, a 21,400× reduction from the random initialization [102]. The autotune + density-filter run is within a factor of 5 on raw MA loss but delivers a manufacturable near-binary geometry respecting the $R = 5$ feature-size constraint; the MMA run delivers a lower loss but a free-form permittivity distribution that requires post-hoc binarization (which in our experience destroys performance by 2–3 orders of magnitude). The β-continuation result is the first one in this pipeline that produces a design suitable for direct fabrication.

9.7 Feasibility discussion

At the design frequency $\lambda = 1550$ nm, the shot-noise-limited detection threshold we report corresponds to $\sim 64\,\mu$W per port at $1\,\mu$s integration, within the operating range of standard telecom lasers. The final binary permittivity distribution is smooth at the density-filter length scale ($R = 5$ pixels ≈ 200 nm feature size at 5-nm/pixel lattice) and compatible with electron-beam lithography on standard SOI [80, 92].

10 Discussion

10.1 Relation to Fisher-information-bound and wave-scattering programs

The paper's $\Phi_{\log J}$ choice in Case Study C connects to the recent quantum-Fisher-information program on wave scattering [12, 16, 88, 107], which has established theoretical maxima of $\det J_N$ over admissible probe states for specific scattering geometries. Our framework is complementary: those results tell us what $\sup_{\boldsymbol{c}} \det J_N$ is; our framework specifies the joint $(\boldsymbol{c}, \pi)$ that

realizes a given $\log J_N$ value and integrates it with the measurement policy. In the Gaussian-posterior regime the two programs converge to the same object (the log-determinant of the Bayesian FIM), but the quantum-FI literature does not treat the policy π as a first-class optimization variable, and our framework does not assert upper bounds on J_N beyond what passivity / reciprocity already imply. We view the two as complementary and look forward to integration in problems where both an upper bound on J_N and a realization of the policy that saturates it are required.

The photonic information-bound program on scattering-matrix extremization [20, 36, 47, 58, 67, 68, 72, 73, 89, 101] plays a similar complementary role. Those results establish upper bounds on various figures of merit (absorption, scattering, local density of states) achievable by *any* passive $\boldsymbol{c}$; ours tells us which specific $\boldsymbol{c}$ realizes the bound and supplies the measurement policy that consumes it. Case Study C's 90,000-pixel design problem is precisely the regime where these bounds become operationally useful as certificates of near-optimality.

10.2 End-to-end imaging and metaoptics: what this paper adds

End-to-end optics, including the information-theoretic design lines by Waller and collaborators [11, 52, 53, 76] and the differentiable-camera-pipeline work by Wetzstein, Heide, Majumdar, and others [3, 19, 39, 51, 65, 97, 108**?**], co-designs hardware with a *fixed, non-adaptive* one-step reconstruction network. Case Study C is in the same category in the sense that it uses differentiable electromagnetic simulation and topology optimization [23, 32, 48, 60–62, 70, 113], but it differs in one essential structural way: the reconstruction is a multi-step EKF that processes observations *as they arrive* and the sensor chooses measurement parameters adaptively. The joint optimization of physical geometry with a multi-step adaptive reconstruction is the novel piece. In problems where adaptation is not advantageous (shot-noise-limited single-exposure imaging, for example) our framework reduces cleanly to end-to-end optics; in problems where adaptation is available (programmable metasurfaces, reconfigurable photonic mesh networks, tunable filter arrays), it adds capacity that end-to-end optics does not expose.

10.3 Cognitive radar and multi-waveform MIMO

The cognitive radar literature [4, 13, 35, 38] has adopted many ingredients of our framework, POMDP formulations of beam scheduling, MI-based reward, Gaussian-posterior surrogates, but treats the antenna aperture as fixed. Our framework reaches into the aperture, treating element positions and excitations as the outer variables. The radar case study (Section 7) uses a toy beam model to expose the structural mismatch between EVPI-optimal and joint-DP-optimal geometry; integration with a full MIMO aperture model is a natural extension.

10.4 Quantum sensing

Quantum sensing, metrology, and state estimation [14, 15, 17, 30, 34, 78, 82, 87, 106, 107] has a mature framework for characterizing the maximum Fisher information attainable on a given estimation problem. Our Case Study B specializes this to flux-bias sensing in a SQUID-loop transmon [27, 55, 114]. The novel content is the joint optimization with the Bellman-optimal adaptive policy; the extension of classical Kitaev PEA [55] to belief-conditional adaptive phase estimation [7, 33] is well-established in the quantum metrology literature, but the co-optimization with the physical circuit geometry is not.

10.5 What the EVPI *is* still good for

We advocate demoting the EVPI from a design target to a workflow tool. Three concrete roles survive:

1. A cheap pre-screen when the full joint DP is computationally prohibitive. $\mathbb{E}[\mathrm{IG}](\boldsymbol{c})$ requires only V_{oracle} and V_{fixed}, both computable by direct schedule enumeration without backward induction.
2. A post-hoc diagnostic showing how much adaptive headroom is available at the joint-optimum geometry. Small $\mathbb{E}[\mathrm{IG}](\boldsymbol{c}^\star)$ means the joint-optimum is close to the oracle ceiling; large values indicate remaining value from longer horizons, richer action sets, or tighter priors.
3. An interpretive lens on the shape of the $\boldsymbol{c}$-landscape. The non-coincidence of $\arg\max \mathbb{E}[\mathrm{IG}]$ and $\arg\max V_{\text{adaptive}}$ encodes physics: extreme geometries have large ceilings that no policy can reach; interior geometries have smaller ceilings that adaptive policies can consume a larger fraction of.

10.6 Limitations

- **Horizon.** Case Studies A and B are at small horizon ($K = 3, 4$) where exact DP is tractable. At larger horizons the Bellman memoization table blows up, forcing rungs 2–5 of the hierarchy. The Gaussian-posterior EKF surrogate (rung 5) extrapolates to arbitrary K; its accuracy relative to exact DP at moderate K is an open question.
- **Tier-1 $\boldsymbol{c}$.** Case Study B's 7-dimensional geometry is not the full photonic design space. A richer tier-2 geometry including SQUID-loop mutual inductances via analytically differentiable Biot–Savart [64] is architecturally supported and expected to give joint-DP further advantage; this is ongoing work.
- **Finite sample MSE bound.** The Bayesian CRB $1/J_P$ is an asymptotic lower bound. For biased Bayesian estimators on bounded priors at small K the bound is loose by orders of magnitude (Case Study B: $\sim 10^8\times$). Tighter finite-sample bounds such as the Weiss–Weinstein [115], Bobrovsky–Zakai, and Ziv–Zakai [105] bounds would be more informative anchors; integrating them into the framework is a natural next step.
- **Continuous-action Bellman.** The count-statistic memoization exploited in Case Study B breaks for continuous actions. Practical paths include fine discretization (as we did), Gaussian belief approximations (rung 5), or particle-filter DP. A full continuous-action Bellman for the scqubit problem at $K \geq 4$ would require one of these paths; we presented the discretization path as the cheapest.

10.7 Open directions

Integration with photonic information bounds [20, 67, 73] would allow certified near-optimality of joint-DP geometry within a passive / reciprocal family. Integration with quantum-Fisher-information maxima [12, 107] would allow the same for quantum sensors. Extension of the Φ-family to entropic-optimal-transport-based regularization [79] and to risk-sensitive rewards (expectile, CVaR) is a one-line change in the Bellman backup. Deployment on reconfigurable physical hardware (programmable metasurfaces, tunable photonic meshes, flux-tunable transmon arrays) is the natural experimental application.

11 Conclusion

For any sensor whose physical hardware is designed once but whose measurement policy runs for the lifetime of the device, programmable metasurfaces, cognitive radars, frequency-tunable quantum sensors, reconfigurable photonic-integrated meshes, joint optimization of geometry and policy is the minimum principled procedure. The envelope theorem makes the joint optimization tractable; a principled relaxation hierarchy makes it portable across scales; a family of compatible scalar rewards makes it applicable across communities. The Expected Value of Perfect Information, while classical and instructive, is a diagnostic, not a decision tool: its argmax disagrees

with the joint-DP argmax by factors of $\sim 3\times$ in attainable adaptive value. The deployment-aligned comparison of joint-DP and Bayesian-CRB co-design on a superconducting-qubit flux sensor gives the operational headline: 13.4% reduction in Bayesian MSE at $z = +11.5\sigma$ under three explicit engineering preconditions. These preconditions, a rich action grid, a reward aligned with the deployment metric, and a physically realistic geometry box, name the failure modes an ablating practitioner would otherwise miss, and make the adaptive-hardware co-design a disciplined engineering procedure rather than an ad-hoc flourish.

Code and data. The Julia implementation of all three case studies, including the exact Bellman DP with count-statistic memoization (Case Studies A and B), the envelope-theorem-based outer Adam with per-component box-width preconditioning, the MSE-terminal reward (`Bellman.jl`, `Gradient.jl`, `JointOpt.jl`), the joint PCRB baseline (`PCRB.jl`), and the photonic BFIM+EKF surrogate (`SimGeomBroadBand.jl`, `BFIMGaussian.jl`, `PhEnd2End.jl`), is publicly available. Numerical outputs of all tables in Sections 7–9 are serialized and shipped with the source; reproducing any table requires a single Julia invocation.

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